

Estimating finger joint angles by surface EMG signal using feature extraction and transformer-based deep learning model

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ABSTRACT

Human-machine interfaces frequently use electromyography (EMG) signals. Based on previous work, feature extraction has a great deal of influence on the performance of EMG pattern recognition. Furthermore, the Deep Learning method is supposed to increase performance and not depend on feature engineering. However, directly processing raw signals will require a higher computation rate. This study proposed a new method that combines feature extraction and Deep Learning to address those issues while improving performance, reducing architecture size, and producing a more representative output. The proposed architecture employs the Transformer model as the backbone to get the correlation between elements and focus on the important information for estimating the flexion-extension of finger joint angles. This study uses experiment three of the NinaPro (Non-Invasive Adaptive Hand Prosthetics) DB5 dataset. Each experiment produces 16 Surface EMG data streams from two Myo Armbands devices representing 22 finger joint angles as output. This study compares the windowing process, feature extraction, execution time, and results with previous studies. The results show that the proposed model outperforms the previous study, from 0.957 in the previous study to become 0.970 in this study for the R-Square score. This result is obtained using 100 data points for the windowing process and Median Frequency for the best feature extraction method.

1. Introduction

The electric impulses generated during muscle contraction, known as electromyography (EMG), carry information and represent the subject's movement intentions. EMG is the sum of the Motor Unit Action Potential (MUAP) plus noise, and it can be split into motor units (MU), which are the smallest unit of human muscle [1]. Recently, the EMG signal became famous biomedical input due to its potential for movement generated. The advantages of using the EMG signals, compared to other bio-signals, are the ease of capturing its signal [2], less influenced by any human activity such as eye movement [3], and easier to extract the information from the EMG signals with fewer dimension data [4]. Therefore, it draws attention from many researchers to develop medical applications to identify neuromuscular disorders using EMG signals [5]. The EMG signals are also used as biomedical control for multimedia tools, therapy robotics [6], wheelchairs [7], and many more. The vital step for leveraging EMG signals as a powerful tool is extracting features from

them and using it as helpful information. Many biomedical applications are developed using signal processing and Artificial Intelligence (AI) techniques [8].

There are two types of tasks in EMG signal processing: classification and regression. An example of the classification task is processing and correlating a raw signal to specific hand gestures using AI [1]. For instance, Anam et al. [9] conducted the work to classify 12 fingers' movements by leveraging the combination of feature extraction and Machine Learning. That work proposed a new feature extraction set consisting of Slopes Sign Change (SSC), Zero Crossing (ZC), Hjorth time-domain parameters (HT), and other time and frequency domain feature extraction. This proposed feature extraction was then classified using Radial Basis Function Extreme Learning Machine (RBF-ELM). Khushaba et al. [10] developed a Deep Learning method by designing fixed convolution-based time-domain feature extraction to discriminate 53 movements from the NinaPro database 5. In another work, William et al. [11] explored classifying specific movements for arm, elbow, arm out,

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and arm up with 90-degree movement by the Recurrent Convolutional Neural Networks (RCNN) approach. However, the given drawback prevented the system from having intuitive control and acting naturally.

Therefore, the recent progress is toward developing the intuitive control and usability of such a human-machine interface, which replaces the classical classification scheme with a regression scheme [8]. Following this scheme means that the system will not distinguish between predetermined gestures but will predict the angle of each joint. As a result, the user can, ideally, freely move the hand while the system still accurately predicts its movement. Therefore, its advantages will benefit such applications as robotic hand therapy for stroke survivors [12,13]. It is believed that the electric signal produced by EMG corresponds to muscle activity, as described by the strength of hand power and movements such as finger grasping and flexion/extension movements.

Conventional therapy for stroke patients that the therapist controls has the disadvantage that it is difficult to evaluate the patient's strength during therapy. This disadvantage may cause inconvenience to the user, such as excessive pain. Hand robots overcome such problems by assessing each finger's movements and adjusting the joint angles of each stroke patient's finger. The therapy process can be carried out independently and monitored using bio-signals. In this way, the user can determine their strength at a specific joint angle according to the therapist's suggestion [14,15]. This bio-signal-based therapy method is also useful for helping post-stroke recovery naturally by restoring damaged brain parts through direct human signals, the most important components in the rehabilitation process [16]. This therapy method uses a specific signal processing method that generalizes the EMG signals and correlates them with specific joint angles of fingers. Therefore, to contribute to this intuitive control in the EMG field, especially for fingers, we designated a signal processing method that leveraged a combination of feature engineering and Deep Learning. Using this approach, we expected to reduce computation costs and accurately predict the finger joint angles. The result will be compared to our previous work and state-of-the-art for determining our contribution.

For reporting our work, this paper is organized in the following. [Section 2](#) presents the literature review of the state-of-the-art EMG regression case. [Section 3](#) presents the experimental setup, including the dataset and the various methods. [Section 4](#) provides the method to test the proposed model. [Section 5](#) presents the results, discusses them, and compares them with our previous work [8] and other similar work to demonstrate our contribution. [Section 5](#) shows the limitations and future work regarding our study. Finally, [Section 6](#) is the conclusion.

2. Literature review

In EMG signal processing, various approaches are mostly based on traditional Machine Learning or shallow learning, such as KNN, SVM, or Linear Regression. Another approach is using Deep Learning like CNN and LSTM or combining traditional Machine Learning and Deep Learning [17]. For instance, the experiment conducted by Alazrai et al. [18] successfully estimated 22 finger joint angles by leveraging Discrete Wavelet Transform (DWT) and Support Vector Regression (SVR) which performed 12 movements. This model achieved an average R-Square score above 60%. Anam et al. [19] have done the simultaneous and proportional myoelectric control systems (MCS) approach using time-domain feature extraction and random forest techniques to estimate 14 finger joint angles. Experimental results indicated that Root Mean Square (RMS) was the most useful feature when combined with Random Forest. This study gets the R-Square score of 0.85. Wang et al. [20] estimate 20 joint angles from six grasp movements by leveraging Long Short-Term Memory (LSTM) as regressor and Mean Absolute Value as feature extraction. Using Pearson Correlation Coefficient (CC) to evaluate the model, the author got 84.02%. Batayneh et al. [21] adopted the same task as the previous researcher who used the NinaPro dataset experiment 3. However, 15 joint angles were tried to be estimated using Root Mean Square (RMS) as feature extraction and Generalized

Regression Neural Network (GRNN) as a regressor in this study. The mean Correlation Coefficient (CC) was applied to validate the result and achieved 85.97%. Geng et al. [22] made a convolution model with an attention mechanism (CNN-Attention) to regress ten channels of finger joint angles with an average R-Square score of 0.73. Anam et al. [9] estimate eight finger joint angles using a combination of LSTM as a regressor, MAV and RMS as the feature extraction, and Principal Component Analysis (PCA) as the dimension reduction. The proposed model reached 87.4% in the R-Square parameter. In another work, Anam et al. [23] developed a model for regressing eight finger joint angles on 24 movements using deep neural network architecture. In this work, they used Mean Absolute Value (MAV), Waveform Length (WL), Integrated Absolute Value (IAV), Zero Crossings (ZC), and Slope Sign Change (SSC). They achieved an average result of 95% on the R-Square score. Guo et al. [24] estimate 22 joint angles using the same model as the previous LSTM. The researcher evaluated the model using Pearson Correlation Coefficient (PCC) and achieved 90.62% accuracy representation. Ilyas et al. [25] successfully combined feature extraction, dimensional reduction, and regressor to calculate the finger joint angles. In this work, they explored Root Mean Square (RMS) and Mean Absolute Value (MAV) for feature extraction, Linear Discriminant Analysis (LDA), Principal Component Analysis (PCA), and Non-Negative Matrix Factorization (NMF) as dimensional reduction and Gated Recurrent Unit (GRU) as its regressor. As a result, it achieved an R-Square score of 0.921 from 52 movements in ten healthy subjects. The last research is our research [8], exploring six types of feature extraction, dimensional reduction, and Machine Learning. We achieved the best result by deploying LSTM with an Attention Mechanism with 0.957 in the R-Square score. The robustness of the attention mechanism lies in its algorithm. The attention mechanism can correlate importance values with the target, allocated with different weights to contribute more to regression performance.

Based on the literature review, the performance of EMG pattern recognition heavily depends on the choice of feature extraction [9]. The feature extraction will help the model process the EMG signal faster by transforming the raw EMG signal into specific information. Therefore, extensive attention is used to designing feature extraction methods, known as feature engineering, that need a human expert to build [26]. With this concept, the computational cost will be reduced significantly.

On the other hand, traditional Machine Learning or shallow learning, such as KNN, SVM, or Linear Regression, is hard to generalize a complex feature space and highly depends on featuring engineering. With this issue arising, the Deep Learning method was introduced. This method has repeatedly proven to automatically acquire knowledge by processing large amounts of unprocessed data, preventing manual intervention when building highly sophisticated systems. Some important benefits from this development include capturing complex patterns and interdependence in vast amounts of information that traditional Machine Learning approaches may not easily comprehend [27]. Despite that, it brought a drawback to massive computation due to the high dimension of raw EMG signal processing in Deep Learning architecture [28]. The massive weight created during the training process affected the model's size and training time [8], making this approach hardly deployed in edge devices.

Accordingly, it is essential to have a signal processing mechanism that reduces dimensions while still capturing information accurately to improve performance. Therefore, in this work, we proposed a model to counter those problems by combining two approaches: feature engineering and the Deep Learning model. Feature engineering extracts the raw EMG signal to become a specific information signal and reduces its size. For this part, we chose two dominant feature types: time domain and frequency domain, with six numbers of feature methods that will be further investigated. Those feature extraction types will be combined with our proposed regressor based on deep learning approaches. By pairing deep learning with the feature extraction method, it is expected to enable our model to perform a double feature crafting mechanism and

create new feature sets from the representative input signal [29]. This mechanism will also reduce the architecture size of Deep Learning and the computational cost [30].

Additionally, we tailored our proposed model to enable it to mine and concentrate on the essential part from extracted signals to anticipate performance reductions as feature space is reduced. Hence, we will use multi-head-attention architecture, called Transformer [31], to keep and organize that information. The Transformer model stands out for its exceptional capability to allocate diverse weights or attention scores to every token in a sequence based on their relevance to other features. This attention mechanism allows focusing on vital features necessary for accomplishing a particular task. Following this multi-head-attention mechanism, multiple neural network layers process its output. This layer transforms the higher-dimensional space generated by the attention layer to correlate it to the expected output, known as the joint finger angles. This research contributes to crafting a robust model that can estimate flexion-extension of finger joint angles, perform better, and reduce the computational cost, especially in our previous work [8]. In addition, this work also investigated some of the parameters of the windowing process, feature extraction, and the regressor to find the best combination and potential drawbacks when applied.

3. Experimental setup

3.1. Dataset

This research uses part of the NinaPro (Non-Invasive Adaptive Hand Prosthetics) dataset. The NinaPro datasets contain EMG data and finger joint angles divided into some datasets [32]. This study only uses DB 5 from the NinaPro dataset. NinaPro dataset DB 5 used the Double Myo-Armband with a frequency sampling of 200 Hz. It contained the surface EMG (S-EMG) data and 22-finger joint angles movement (extension-flexion signal at the metacarpophalangeal) data. The 22-finger joint angle movement marks are shown in Fig. 1. This study uses five subjects (S1 to S5). For every subject, a set of movements are done and repeated with the right hand. Each movement repetition lasted 5 s and was followed by 3 s of rest. The movement in this study used exercise three (E3), which provides 23 finger movements, as shown in Table 1. From these movements, the 22-finger joint angles will be measured. Using this setup, our proposed model used the S-EMG data to become the input

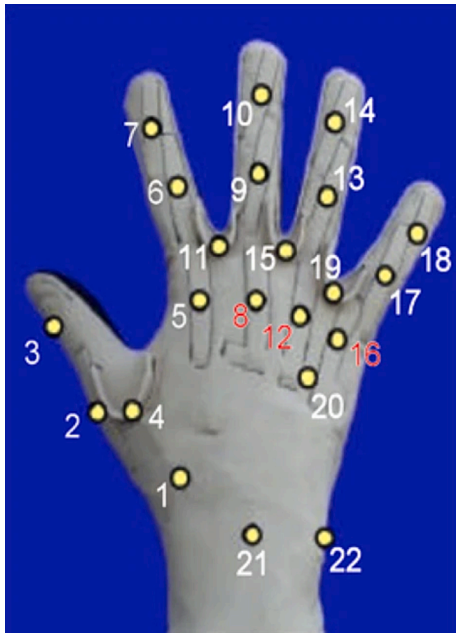


Fig. 1. The 22-finger joint angle marks from CyberGlove II [33].

Table 1

Finger Movements in NinaPro dataset DB 5 Exercise 3.

Number	Movements name	Number	Movements name
1	Small diameter grasp (power grip)	13	Tripod grasp
2	Large diameter grasp	14	Quadpod grasp
3	Precision sphere grasp	15	Ring grasp
4	Fixed hook grasp	16	Tip pinch grasp
5	Prismatic four fingers	17	Parallel extension grasp
6	Medium wrap	18	Lateral grasp
7	Index finger extension grasp	19	Extension type grasp
8	Stick grasp	20	Power disk grasp
9	Writing tripod grasp	21	Open a bottle with a tripod grasp
10	Power sphere grasp	22	Turning a screw
11	Three-finger sphere grasps	23	Cutting something
12	Prismatic pinch grasp		

signal and the 22-finger joint angles movement data from dataglove to become the output to be regressed. In other words, we propose a model to estimate the 22 joint angles from the S-EMG.

Fig. 2 shows the experimental setup of this experiment, including the Double Myo-Armband and the CyberGlove II setup. The Double Myo-Armband gives 16 channels (2x8 channels) that can be accessed to become the model's input. From these 16 channels, we retrieved 16 representative signals. While the CyberGlove II provides 22 channels, representing the finger joint angles, to become the output, as shown in Fig. 1.

3.2. Model setup

As previously stated, this study aims to predict the finger angle (flexion-extension signal at the metacarpophalangeal (MCP) joint). This study's constructed model is based on the encoder transformer model. Generally, the constructed model consists of five blocks, as depicted in Fig. 3. The detailed process of each block is further elaborated in detail in a specific section.

3.2.1. EMG channel

The first block is the input block. This block retrieves the pure signal from the Double Myo-Armbands device with a total input channel of 16 channels (2 Myo $\times$ 8 channels) that gives 16 representative input signals. The devices used 200 Hz frequency sampling, giving 200×16 data for one second. Therefore, this signal would then be further processed.

3.2.2. Pre-processing and windowing

The second block of this model is the filtering and windowing section. This section performs pre-processing in the form of filtering and denoising of the input EMG signal. In addition, the windowing process (sliding window) is also employed in this section.

This windowing process is where a fixed-sized window is moved or slid over the whole signal sequence to get a shorter signal, dividing the whole sequence signal into smaller groups for the next process step. This study did not use an overlapping window on the windowing process to make it simpler and perform faster with fewer windows. The unit used for data windowing is the data point. The lower the value, the less data is processed at one time, and vice versa. For example, 100 data points mean 100 data will be obtained and further processed at one time. This windowing process will significantly affect the accuracy and processing time produced by the built model [34]. This study uses four values of the windowing process; 25, 50, 100, and 200 data points [35]. After windowing the data, the following process removes the low and high-frequency noise in the S-EMG signal using a Bandpass filter. The band-pass filter value is set into the range of 15–25 Hz. This value is based on the bandwidth frequency in the raw S-EMG signal and the processed S-EMG signals obtained from the electrodes [36].

To provide more representative features or information for the whole

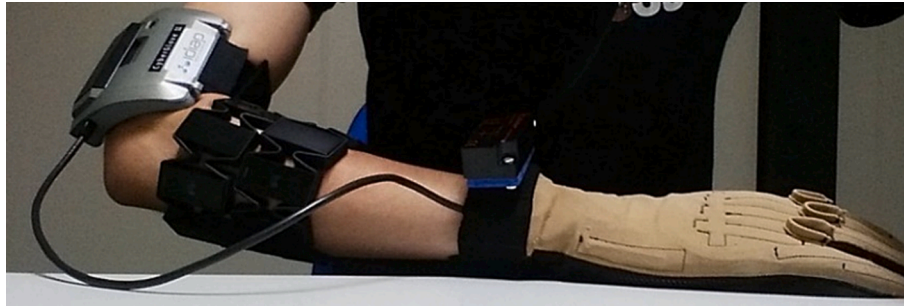


Fig. 2. Experiment setup using Double Myo-Armband and the CyberGlove II [32].

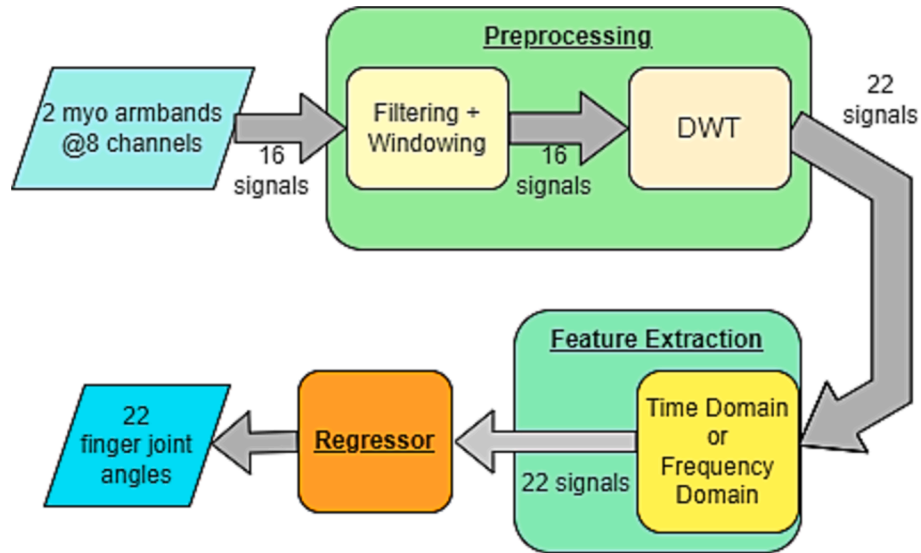


Fig. 3. The model consists of five blocks: EMG Channel as the input, Pre-processing and Windowing, Feature Extraction, Regressor, and 22-Finger Joint Angles as the output.

scheme by transforming it from time to frequency domain, thus, we also involve Discrete Wavelet Transform (DWT) [37]. DWT provides a structured multi-resolution analysis and improved frequency localization. DWT allows decomposing the original EEG signal into narrow-band signals, offering a detailed view of the signal's frequency content at different scales. Thus, DWT can improve the system over methods like Empirical Mode Decomposition (EMD), which can suffer from

limitations, such as imperfect decomposition in the low-frequency region and wide frequency bands of the obtained intrinsic mode functions (IMFs), and give better denoising effect [38,39].

There are two coefficients, the Detail (D) coefficients give higher frequency and are resolved over time, and the Approximation (A) coefficients give lower frequency that achieves superior frequency resolution [40]. The DWT method runs by applying the decomposition

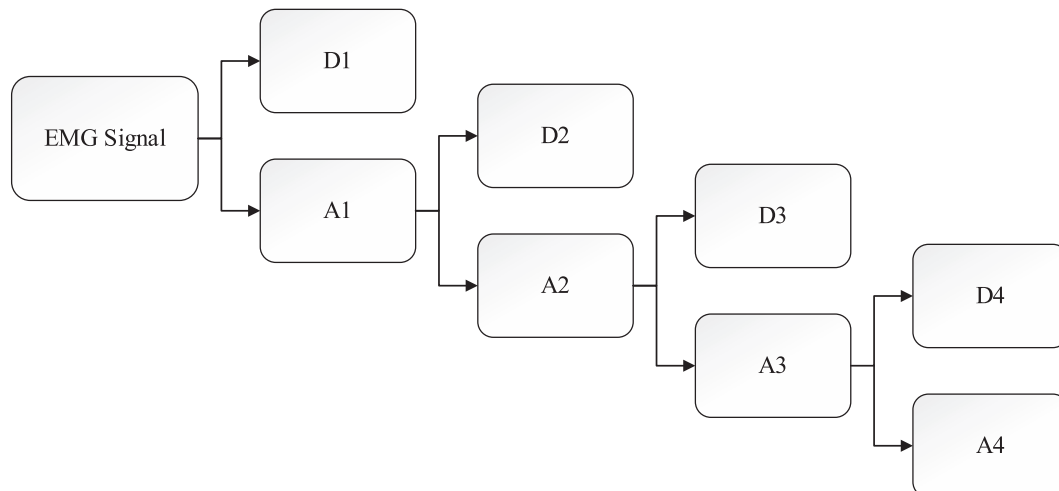


Fig. 4. Discrete Wavelet Transformation Tree for D4 and A4.

principle. In each stage of the process, an input signal (coefficient) will be divided into two new signal values (coefficients), namely the Detail coefficient (D) and the Approximation coefficient (A). In the following process, from one of the new signals (coefficients) obtained earlier (coefficient A), the decomposition process will be carried out again into the new Detail (D) and Approximation (A) coefficients. A similar process will be carried out again for the stages at the next level. DWT increases the frequency resolution and reduces the sample count by doubling the frequency resolution and reducing the sample size. For more details, Fig. 4 shows detailed information about the DWT tree for the wavelet used, with D1 as Detail coefficient level 1, A1 as Approximation coefficient level 1, D2 as Detail coefficient level 2, A2 as Approximation coefficient level 2, D3 as Detail coefficient level 3, A3 as Approximation coefficient level 3, D4 as Detail coefficient level 4, and A4 as Approximation coefficient level 4.

Based on previous research, DWT with four decomposition levels can reduce noise from EMG signals or other biomedical signals [37,41]. In addition, DWT with four levels of decomposition is also believed to help the pattern recognition process for EMG signals [41]. Based on this information, this study also uses four levels of DWT decomposition. Of the four levels of decomposition that have been carried out, the coefficients D4 and A4 are believed to provide the best results [41]. Therefore, the research will use the two coefficients. From the two coefficients, 22 new signals are obtained, combining 11 signals from D4 and 11 other signals from A4.

3.2.3. Feature extraction

The third part of this model is feature extraction. Feature extraction aims to obtain new information from specific characteristics of the original input signal with certain methods. There are three types of feature extraction in EMG-based signal processing: time domain, frequency domain, and time-frequency domain [42]. However, the time-frequency domain has a drawback because of its time processing [43]. Therefore, this study employs two forms of feature extraction: time domain and frequency domain. Time domain features consist of Mean Absolute Value (MAV), Variance (VAR), and Root Mean Square (RMS). Frequency domain features consist of Mean Power (MNP), Mean Frequency (MNF), and median frequency (MDF). Every feature on every domain will be used separately. After this step, the proposed model will get 22 signals containing one of these six features. The effect of each feature will be compared to see which extraction method is suitable for this study.

As for the time domain features, Mean Absolute Value (MAV), Variance (VAR), and Root Mean Square (RMS), these three types of feature extraction are widely used in EMG-based signal processing because they can give features that represent a specific characteristic of EMG-based signal [44]. Mean Absolute value is one of the most popular features used in EMG-based signal analysis [45]. Other names sometimes call this feature; Integral of Absolute Value (IAV), the first order of v-Order features (V1), Averaged Absolute Value (AAV), and Average Rectified Value (ARV). This feature is an average absolute value of the EMG signal amplitude in a segment. The mathematical definition of this MAV feature is shown in Eq. (1).

$$MAV = \frac{1}{N} \sum_{i=1}^N |x_i| \quad (1)$$

The variance of EMG, or Variance (VAR), is generally defined as an average of square values of the deviation of that variable. However, the mean value of the EMG signal is close to zero ($\sim 10^{-10}$) [45]. So, the mathematical definition of VAR can be defined in Eq. (2).

$$VAR = \frac{1}{N} \sum_{i=1}^N x_i^2 \quad (2)$$

The last feature from the time domain group is Root Mean Square (RMS). RMS is another popular feature for analyzing the EMG-based

signal. This feature is like a standard deviation method. RMS is modeled as amplitude modulated gaussian random process that relates to constant force and non-fatiguing contraction [45]. The definition of the RMS equation can be expressed in Eq. (3).

$$RMS = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2} \quad (3)$$

For the frequency domain, Mean Power (MNP), Mean Frequency (MNF), and median frequency (MDF) are regarded as suitable feature extraction methods in EMG-based signal processing [46]. The first one is MNP. The MNP is based on the power of the frequency value. This feature calculates the average power of the EMG power spectrum [45]. The second feature is MNF, defined as the sum of the products of the EMG power spectrum and the frequency divided by the total number of power spectrums. The last one is MDF. MDF is the frequency at which the EMG power spectrum is divided into two regions of the same integrated power [47]. The mathematical definition of those feature extraction was shown in Eqs. (4) to (6).

$$MNP = \sum_{j=1}^M \frac{P_j}{M} \quad (4)$$

$$MNF = \frac{\sum_{j=1}^M f_j P_j}{\sum_{j=1}^M P_j} \quad (5)$$

$$MDF = \frac{1}{2} \sum_{j=1}^M P_j \quad (6)$$

where P_j is the EMG power spectrum at a frequency bin j , f_j is the frequency of the spectrum at a frequency bin j , and M is the total number of frequency bins.

3.2.4. Regressor

The last part of the model is the regressor block. The regressor in this model is used to estimate the 22 values of the flexion-extensions of finger joint angles from the processed signal passed by the previous block. This study mainly used a transformer-based regressor to calculate the output. Fig. 5 shows the architecture of the transformer model.

In Fig. 5, the transformer architecture employed comprises several processing steps. The first is the transformer block. This transformer block consists of layer normalization, multi-head attention, dropout, and feed-forward (FF). A convolutional layer is put inside the FF part with 'relu' as an activation function. It is intended to learn representative features from the input after it has been processed by multi-head attention to identifying significant areas of interest. Before the regression block is attached, this transformer block stacks into an n-number. This regression block is intended to learn the global feature and separate it into several classes based on it. The regression block consists of global average pooling to summarize the produced feature space and MLP with 'relu' as the activation number. All parameters used in the proposed architecture are presented in Table 2, where each layer's order is related to how close it is to the input layer.

3.2.5. Finger joint angles

The last part of the diagram block is the output block. As previously stated, the output of this model is the 22-finger joint angles. Furthermore, the proposed model estimates each angle's values for the 22-finger joint angles using the previous steps: pre-processing and windowing process (sliding window), feature extraction, and regression process.

3.3. Testing scenario

This research aims to determine the most appropriate combination of

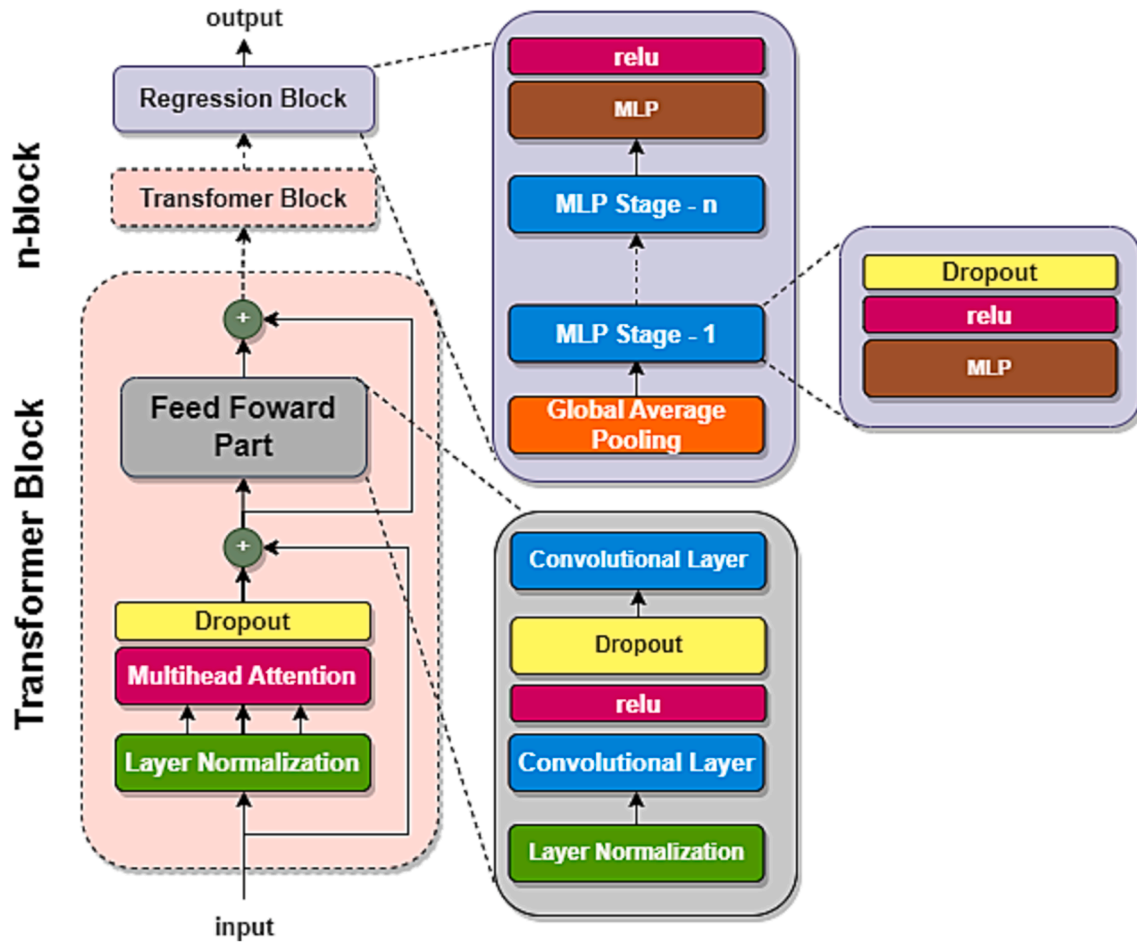


Fig. 5. The Transformer Architecture.

Table 2
Parameters Used in the Proposed Transformer Architecture.

Layer	Parameter	Value
Transformer Block	Number of transformer blocks	4
	Head size	256
	Number of head	4
	Normalization epsilon	1e-6
	Dropout	0
Feed Forward Part	Normalization epsilon	1e-6
	1st Convolutional filters, kernel size, and activation	3, 1, 'relu'
	Dropout	0.25
	2nd Convolutional filters and kernel size	Number of channel inputs, 1
Regressions Block	1st MLP units, activation	128, 'relu'
	MLP dropout	0.4
	2nd MLP units, activation	22, 'relu'

the number of data points for the windowing process, feature extraction, and regressor for estimating the flexion-extension of the finger angle. The variety of this experiment will have 24 designated models as the effect of individual pairing, which consists of four variation numbers of the data point for the windowing process, six feature extraction, and one regressor.

We used the K-Fold method with $K = 5$ to evaluate the model. Each result data will be saved into the data logger during testing. Each pair was compared to check the best performance that could be achieved.

Finally, the experimental result from this study is also compared to the previous studies with similar purposes. This study uses statistical parameters to test the model; they are R-Square (R^2), Mean Square Error (MSE), and Root Mean Square Error (RMSE). Those parameters are used to evaluate a regression task because they represent the model's accuracy and performance [48].

The first parameter is R-Square. R-square is used to find the similarity of the finger's flexion-extension angle value estimated by the model and the real one [48]. The higher the R-square value, the better the model. The mathematical definition of the R-square is shown in Eq. (7).

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \quad (7)$$

where i is the number of the data, y_i is the i^{th} data, $\hat{y}$ is the predicted data, and $\bar{y}$ is the mean value.

The second parameter used is the Mean Square Error (MSE). The MSE parameter determines the error between the predicted and the actual values. It also describes the level of loss associated with the results [48]. For the model to become good, the MSE value should be small. The mathematical definition of MSE can be seen in Eq. (8).

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (8)$$

The last parameter used in this study is Root Mean Square (RMSE). This parameter is used to evaluate the numerical error of amplitude [20]. For the model to become better, the RMSE value should be smaller. The RMSE mathematical definition can be shown as follows.

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{n}} \quad (9)$$

This study calculates the MSE and RMSE values for all the output channels as a single output value. Because the output for this study is 22 finger joint angles, the MSE and RMSE will be used to calculate the 22 angles as one output value.

For evaluation, this work was set up with Python programming language, with the following machine specifications to assess performance: Intel(R) Xeon(R) CPU E5-2630 v4 @ 2.20 GHz, RAM 98, and boosted with GPU NVIDIA RTX-2080 Ti. Then, we used Adam and MSE with a learning rate of 10^{-5} for training parameters.

4. Results and discussion

System testing is based on the model described earlier in [section 2](#). Each test uses K-Fold cross-validation with a K value of 5. This testing uses all existing training data to provide a model with a more negligible bias. The resulting data is taken by storing the required parameter values in a data logger. Then the analysis is carried out afterward. The discussion of the study of the results will be divided into several sections to make it easier to see the effect of each parameter being tested.

4.1. Windowing process (Sliding Window) evaluation

This section will discuss the effect of the windowing process on the architectural model being built. The data points of the study's window are 25, 50, 100, and 200. The windowing process will affect the amount of data processed in one process. In the end, this will affect the processing time.

For example, in this study, the EMG sensors used were two Myo Armbands, each having eight channels with a sampling frequency of 200 Hz. This 200 Hz is the default frequency sampling produced by Myo Armband. Therefore, it will get 200×16 sensor data in one second. Suppose the model uses 200 data points for the windowing process. In that case, the model must wait one second to do one data processing with a size of 200×16 . It is different if the windowing process used 100 data points, then the model only takes half a second to do a one-time process with a data size of 100×16 . Likewise, for 50 and 25 data points, each only takes 0.25 s for data of 50×16 and 0.125 s for data of 25×16 . Thus, the smaller the data points for the windowing process, the faster the model can produce predictions. This test determines the most appropriate number of data points for the windowing process for the model being built.

The test results are presented in [Table 3](#). The experimental results shown in [Table 3](#) are the average testing value in several variations of the feature extraction method employed for each windowing process.

Based on the data obtained, the highest average value of R-Square is obtained for the windowing process with a value of 100 data points, equal to 0.970. The lowest R-Square average value was obtained for the windowing process with 200 data points, with a value of 0.968. The R-Square value obtained increases with the increase in the number of data points, namely at 25 to 100, before finally decreasing for the windowing process with 200 data points. Based on this phenomenon, we can decide that the most appropriate number of data points for the windowing process is 100.

Table 3

Comparison Result in Varying Number of Data Points for the Windowing Process.

Parameter	Windowing Process (Data Points)			
	25	50	100	200
Mean R-Square	0.969	0.969	0.970	0.968
Mean MSE	960.121	965.812	970.888	1003.302
Mean RMSE (⁰)	20.716	20.976	21.104	21.511

On the other hand, we can also see that the higher the number of data points used, the higher the average MSE and RMSE values get. The lowest MSE value was obtained in the windowing process using 25 data points, 960.121. In comparison, the highest value was obtained for the windowing process using 200 data points, namely 1003.302. The same thing happened to the average RMSE values obtained. The lowest RMSE value is 20.716⁰ for the windowing process using 25 data points, and the highest RMSE value is 21.511⁰ for the windowing process using 200 data points.

Contrary to the R-Square value, the best number of data points for the windowing process is done when the MSE and RMSE values are small. These MSE and RMSE values are calculated as a single output value for all the output channels. The output for this study is 22 finger joint angles. So, the MSE and RMSE for this study will calculate the error for 22 angles as one value. For example, if the RMSE value is 20.716⁰, this value is for 22 angles or an average of 0.942⁰ for each angle.

It can be concluded that the number of data points for the windowing process needs to be enlarged to get a better R-Square value. However, this will also have side effects in the form of higher MSE and RMSE values. This analysis shows that this experiment's best windowing process uses 100 data points.

4.2. Feature extraction evaluation

This section will discuss the effects of using different feature extraction methods for the developed model. The results obtained are the average values of the tests in several variations of the data point's number for the windowing process that was carried out. This research selected six feature extraction methods, which can be divided into time and frequency domain groups. The time domain group consists of three methods, namely Mean Absolute Value (MAV), Root Mean Square (RMS), and Variance (VAR). In comparison, the frequency domain group consists of Mean Power (MNP), Mean Frequency (MNF), and median frequency (MDF). Every feature on every domain will be used separately. The effect of each feature will be compared to see which extraction method is suitable for this study. The data from testing the effect of using the feature extraction method can be seen in [Table 4](#).

Based on the data above, it can be seen that feature extraction in the frequency domain has quite a similar R-Square result for each method used, which is 0.968. Whereas for the R-Square in the time domain, there is a slight difference in the results obtained. The highest R-Square value for the time domain was obtained using the MDF and MNF methods with the same value of 0.970. While the lowest value was obtained using the MNP method with a value of 0.969. As mentioned earlier, it indicated that higher R-Square values represent more suitable techniques we employed. Thus, the MDF and MNF methods are the best methods to use.

Furthermore, to determine the most suitable method that can be used between MDF and MNF, we can look at the results of MSE and RMSE parameters. As previously explained, in contrast to the R-Square, the smaller the value of MSE and RMSE, the better the model will be. Thus, it appears that the MDF method is superior to the MNF method because the MSE and RMSE values of the MDF method are smaller than the MSE and RMSE values of the MNF method. Even though there are some better values for the MSE and RMSE parameters in other methods, the MDF method is still dominant because it gives the best R-Square value. However, the previous table only shows the quick result of the evaluated windowing process. [Fig. 6](#) details the relationship between each feature extraction and windowing process.

The previous comparison figure shows that the frequency domain as a feature extraction method is higher than the time domain. All feature extraction in the frequency domain reaches 0.97 in every tested data point's number for the windowing process and only has a smaller value to MNP. Time domain feature extraction has a similar result in each selected data point's number for the windowing process. However, when the data point's number for the windowing process gets smaller, it

Table 4
Feature Extraction Domain Comparison.

Parameter	Frequency Domain			Time Domain		
	MDF	MNF	MNP	RMS	VAR	MAV
R-Square	0.970	0.970	0.969	0.968	0.968	0.968
MSE	996.144	1042.069	961.108	942.790	963.064	945.010
RMSE (%)	22.544	23.412	20.209	19.990	20.288	20.018

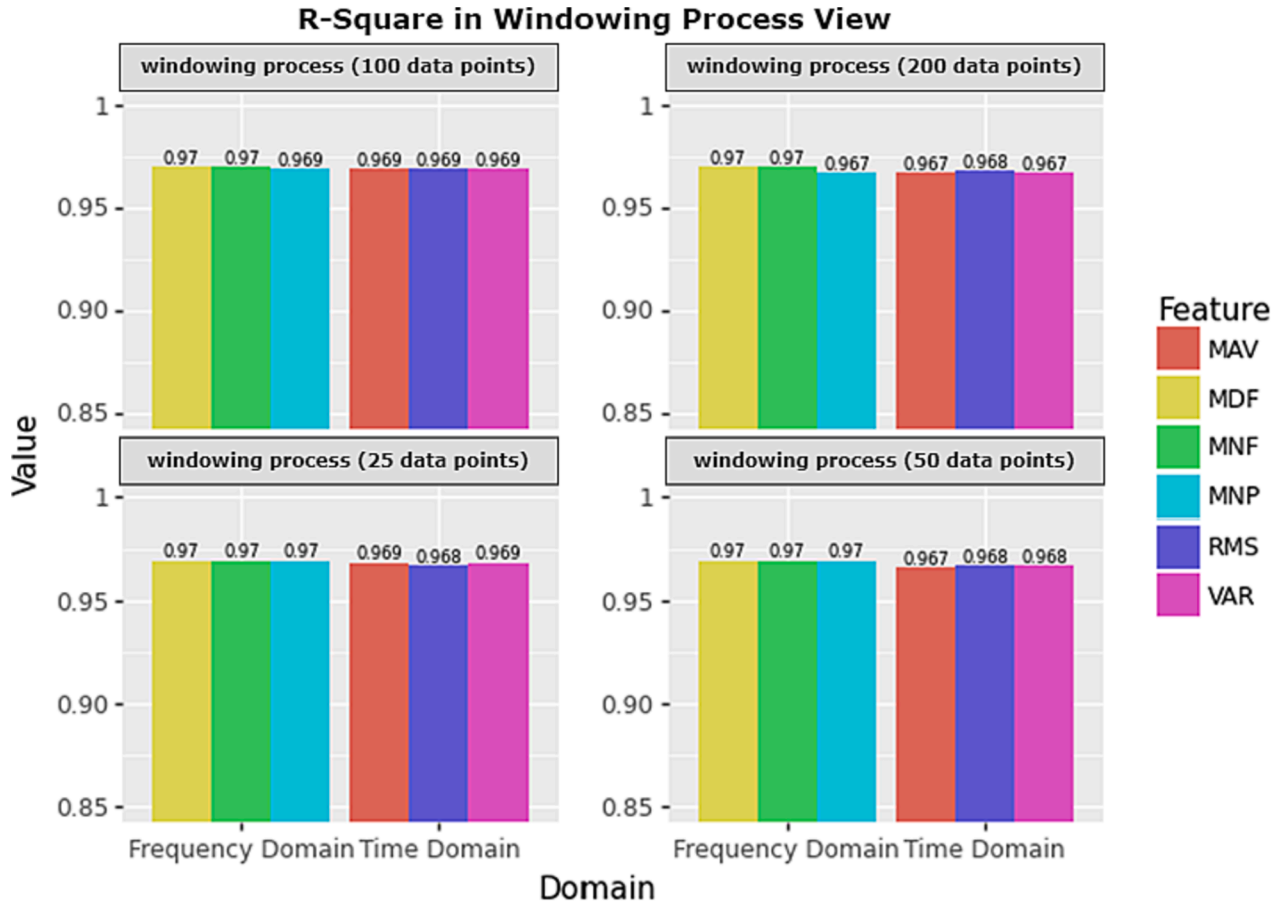


Fig. 6. Relationship Between each Feature Extraction and Windowing Process.

has a slightly small consistent value. It differs from the frequency domain, which increases the R-Square value when the data point's number for the windowing process gets smaller. This phenomenon might indicate that the signal information represents each finger joint angle in a small- data point's number for the windowing process. However, it also gives other probability that the Transformer might need big data to recognize this finger joint angle's pattern. However, this condition might be suitable for the frequency domain compared to the time domain because, in the time domain, there is inconsistent value change whether a smaller or higher data point's number for the

windowing process is applied.

4.3. Execution time evaluation

We capture all information concerning the training phase to evaluate the time matter. This part also becomes important since LSTM, combined with Attention expenses and a lot of computation cost in our previous experiment, becomes a drawback compared to another method. This evaluation will also give an insight into the complexity of data after feature extraction processing is done and computed using

Table 5
The Average Execution Time Comparison until The Model Convergence in milliseconds (ms).

Windowing Process (Data Points)	Time Domain			Frequency Domain		
	MDF	MNF	MNP	RMS	VAR	MAV
25	11946.881	11893.725	3940.118	3752.657	3837.927	3510.924
50	6364.041	7689.856	2512.895	3361.019	3304.211	3276.873
100	4323.440	5331.914	2486.291	3360.037	2867.185	3008.718
200	3663.336	4092.384	2711.381	3226.402	3059.295	3015.657
Total	26297.7	29007.88	11650.69	13700.12	13068.62	12812.17

Transformer. Therefore, in Table 5, we reported our time validation. The validation reported the average execution time for every combination experiment until the Transformer convergence. It used early stopping with patience and stopping parameter 500 epoch in milliseconds. One example of a loss graph formed during the training process can be seen in Fig. 7.

Table 5 shows the average execution time comparison until the model convergence in milliseconds. It shows that the more data points used in the windowing process, the more processing time is needed. Regarding training time for each feature extraction method, the MNP became the fastest feature extraction, followed by MAV, VAR, RMS, MDF, and MNF. This result means that MNP can help the Transformers model to converge faster during training. For more detailed information, we also illustrated the previous evaluation in the form of the figure, as shown in Fig. 8.

Concerning the comparison figure, it clearly shows that MNP looks to have faster convergence than other methods. However, in the windowing process using 100 data points for MNP, MAV, and VAR, they demonstrate to become the optimal number of the data point for the windowing process since they acquired a faster time to make Transformer converge. Further analysis also has been done to evaluate the average execution time per step or each iteration for every combination of experiment parameters. This step is required for a fairer analysis of our previous work to know which method has a low computation cost. All of this information is reported in Table 6 and Fig. 9.

The previous result showed that in terms of the feature extraction domain, the frequency domain has a faster time step compared to the time domain, with MNF becoming the fastest method. Fig. 9 also shows that phenomenon too. As same from the previous result also demonstrated that the windowing process using 100 data points looks to become the optimal choice since it requires less time to calculate in each

iteration.

4.4. Comparison with the previous studies

This chapter will compare our study and the previous studies to see valuable insight into the performance of these models. Four previous studies will be compared in this chapter. These studies are previous studies with the closest topic and goal of estimating finger joint angles using s-EMG signals. Notably, the comparison highlights several differences between these models, resulting in differences in performance scores. The comparison is shown in Table 7.

This table compares the research studies based on their input, output, method used, and the R-Square score achieved. The input refers to the number of channels used from a specific dataset. At the same time, the output represents the number of channels related to the metacarpophalangeal (MCP) joint. The R-Square score is used to assess the performance of the models. The scores range from 0 to 1, with higher values indicating better performance. Furthermore, we add RMSE results from these studies for additional comparison.

In the input aspect, [22] uses 12 channels from the DB2 NinaPro dataset, while [8,9,25] and our study utilizes 16 channels from the DB5 NinaPro dataset. The increase in input channels in the latter studies suggests a more comprehensive capture of surface electromyography (EMG) signals, potentially leading to a richer representation of finger movements.

Regarding the output, [9,22] achieve estimations of 8 and 10 channels, respectively. However, [8,25], and our proposed study demonstrate advancements by successfully estimating 22 channels in the MCP joint. This improvement in output suggests a more detailed estimation of finger joint angles, potentially leading to a more accurate representation of finger movements. Conversely, more outputs channel means harder

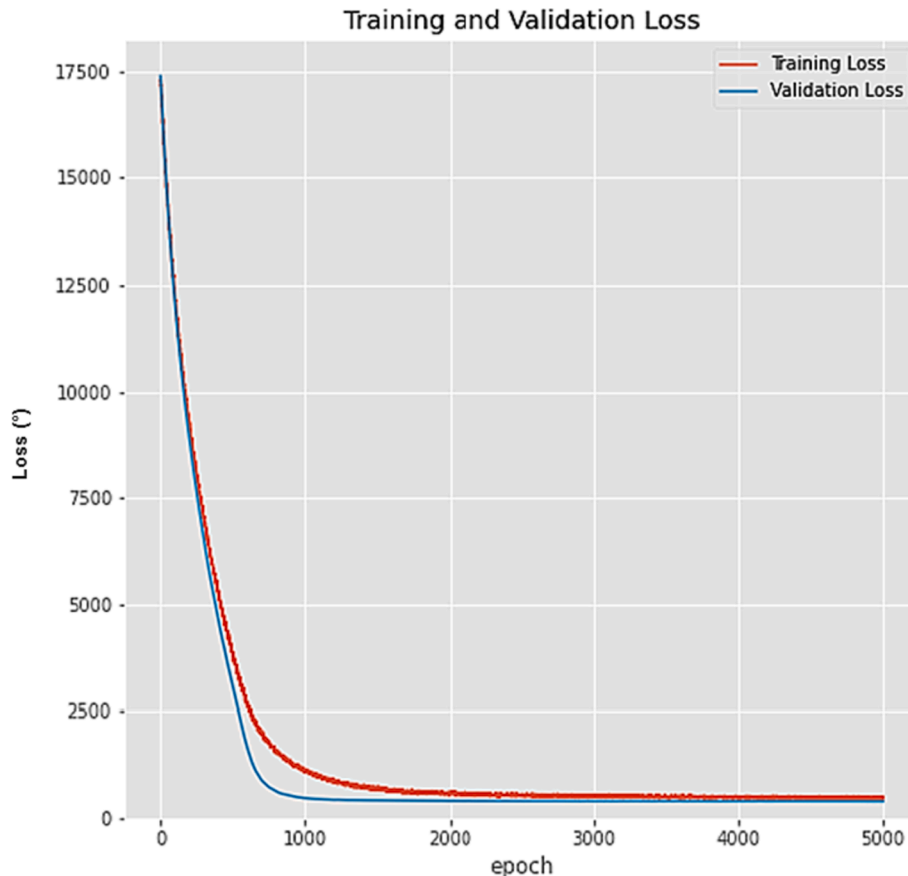


Fig. 7. Loss plot during the training process.

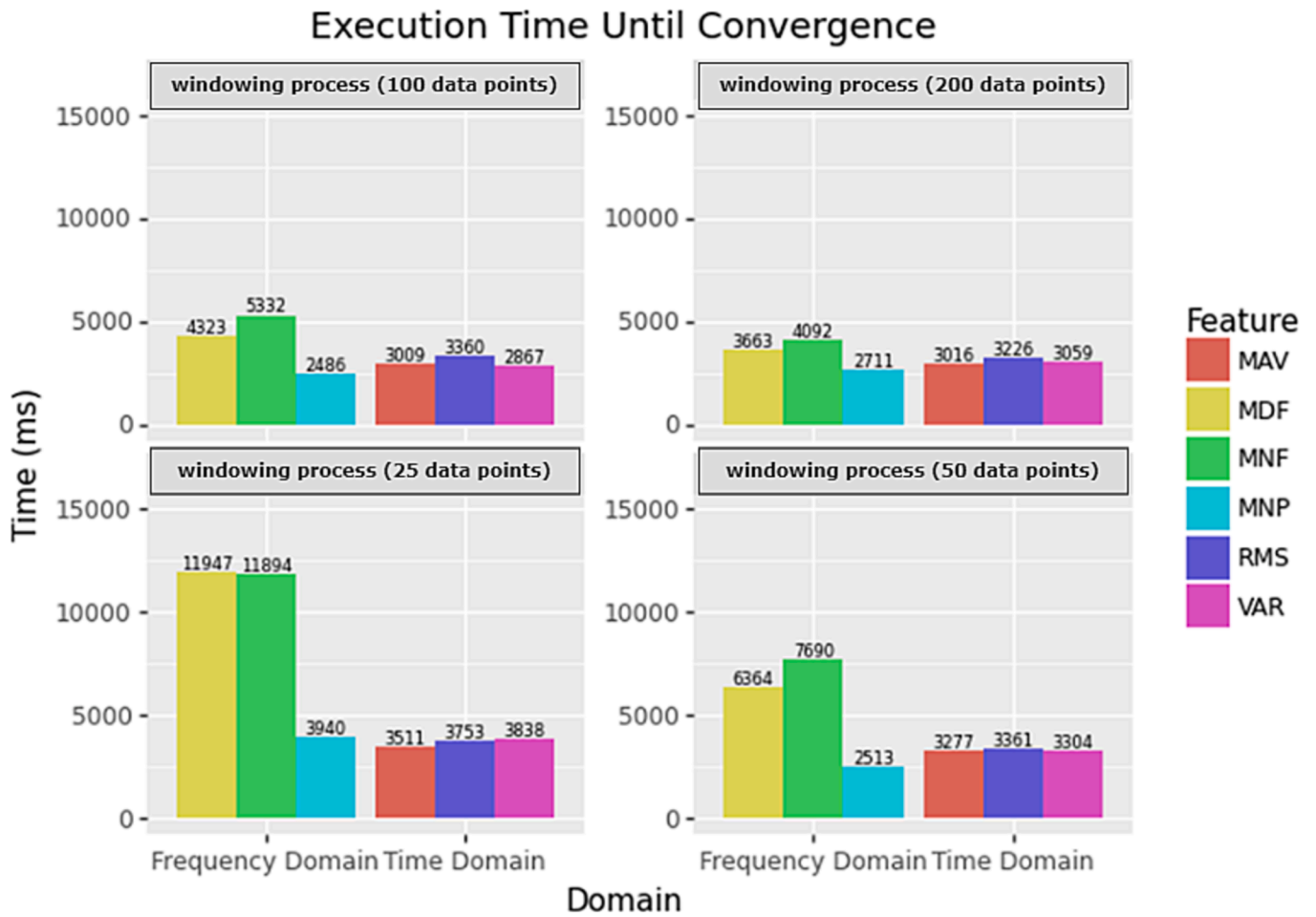


Fig. 8. The Average Execution Time Comparison until The Model Convergence in milliseconds (ms).

Table 6

Average execution time per step for every combination experiment in milliseconds (ms).

Windowing Process (Data Points)	Time Domain			Frequency Domain		
	MDF	MNF	MNP	RMS	VAR	MAV
25	3054.2	3053.2	3054.2	3257.4	3054.8	3254.0
50	2065.6	2061.6	2066.0	2064.6	2068.6	2061.8
100	1075.4	1076.0	1081.6	1083.0	1074.8	1070.2
200	1105.6	1106.2	1105.0	1108.2	1102.8	1101.2
Total	7300.8	7297	7306.8	7513.2	7301	7487.2

for the method to regress the angles.

Regarding R-Square, even with fewer output channels, [9,22] have lower R-Square values than others, including our proposed method. [8,25] have the same input and output as our study, but our proposed method can achieve better R-Square than both. These results indicate that the model presented in this study better fits the data and could better capture the underlying relationships between the input EMG signals and the output finger joint angles. This improvement in the R-Square value could be attributed to using a more powerful model architecture. Therefore, our experimental results presented in this study show that leveraging the transformer network can achieve a higher R-Square value. The results show that the proposed model outperforms the best result from previous studies, from 0.957 in [8], become 0.970 in this study for the R-Square score.

On the other hand, our proposed method still has a few drawbacks when we see the RMSE values. Our proposed method has a bigger value

than [8] but achieves better than [25]. This phenomenon might happen because [8] used the dimensional reduction that helps to choose the right features to help the model get better RMSE values.

5. Limitation and future work

The limitation of this study is that this study examines one kind of exercise from the DB 5 in the NinaPro dataset. Future studies can further examine another kind of exercise from another DB in the NinaPro dataset or another dataset to see more about the robustness of the proposed model. Another limitation of this study is that the performance of the proposed model was only investigated in a non-real-time setting. Examination of the real-time setting may be used to see the real-time response and performance of the proposed model. Future work also can add the muscle fatigue issue to the studies. Using real-time settings and adding the muscle fatigue issue can make the proposed model performs better in the real-case scenario.

6. Conclusion

All variables used to calculate the finger joint angles' flexion extensions have been examined. The components utilized in this research are based on state-of-the-art EMG signal processing. The observed components are the windowing process (sliding window), feature extraction, execution time, and comparison with the previous similar study. This study attempts to find high-accuracy and rapid estimation methods for solving the problem in real-world scenarios. This study can ensure the proposed method's stability with five K-Fold as the testing method. All findings are saved into the data logger for subsequent

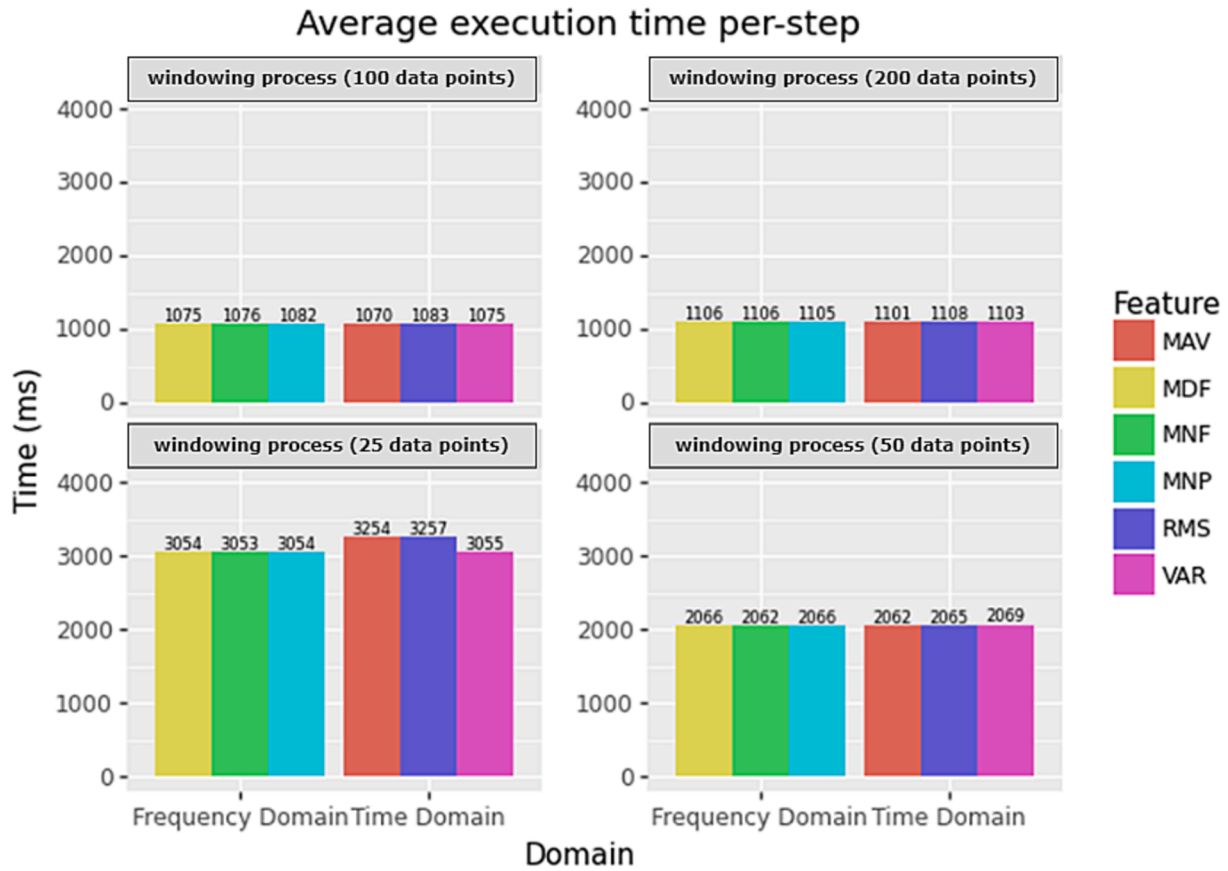


Fig. 9. Average execution time per step for every combination experiment in milliseconds (ms).

Table 7
Comparison with Previous Studies.

Research	Input	Output	Method	Results
A CNN-Attention Network for Continuous Estimation of Finger Kinematics from Surface Electromyography [22]	12 channels from the DB2 NinaPro dataset	Ten channels in the metacarpophalangeal (MCP) joint	Convolution Neural Network with Attention mechanism	$R^2 = 0.730$
Estimation of Finger Joint Movement based on Electromyography Signal using Long Short-Term Memory [9]	16 channels from the DB5 NinaPro dataset	Eight channels in the metacarpophalangeal (MCP) joint	Long Short-Term Memory	$R^2 = 0.874$
Evaluation of Gated-Recurrent Unit for Estimating Finger-Joint Angle using Surface Electromyography Signal [25]	16 channels from the DB5 NinaPro dataset	22 channels in the metacarpophalangeal (MCP) joint	Gated Recurrent Unit	$R^2 = 0.921$ RMSE = 37.110^0
Estimating finger joint angles on surface EMG using Manifold Learning and Long Short-Term Memory with Attention mechanism [8]	16 channels from the DB5 NinaPro dataset	22 channels in the metacarpophalangeal (MCP) joint	Long Short-Term Memory with Attention Mechanism	$R^2 = 0.957$ RMSE = 16.714^0
Estimating Finger Joint Angles by Surface EMG Signal using Feature Extraction and Transformer-based Deep Learning Model (ours)	16 channels from the DB5 NinaPro dataset	22 channels in the metacarpophalangeal (MCP) joint	Transformer based model	$R^2 = 0.970$ RMSE = 21.104^0

analysis.

In this study, it was found that the best window size for the windowing process is using 100 data points. The higher the data points for the windowing process, the higher the average MSE and RMSE values. The Median Frequency (MDF) method proved to be the best method for extracting features from EMG signals in this experiment. In estimating the finger joint angle, the MDF feature extraction and regressor transformer combination give an R-Square value of 0.970 and an RMSE value of 21.104^0 . By utilizing a transformer model, a better R-Square value is obtained. However, there is a slight decrease in the RMSE value compared to our previous study.

CRedit authorship contribution statement

Nur Achmad Sulistyo Putro: Conceptualization, Methodology,

Software, Investigation, Writing – original draft. Cries Avian: Writing – original draft, Visualization, Data curation. Setya Widyanan Prakosa: Writing – review & editing, Validation. Muhammad Izzuddin Mahali: Writing – review & editing. Jenq-Shiou Leu: Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in this study has been described in the article.

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